INCH-POUND
MS27782F
4 May 2024
SUPERSEDING
MS27782E
24 November 2021

DETAIL SPECIFICATION SHEET

SWITCH, TOGGLE, TWO POLE, ENVIRONMENTALLY SEALED, LEVER LOCK, INTEGRATED WIRE TERMINALS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switch described herein shall consist of this specification and the latest issue of MIL-DTL-3950.

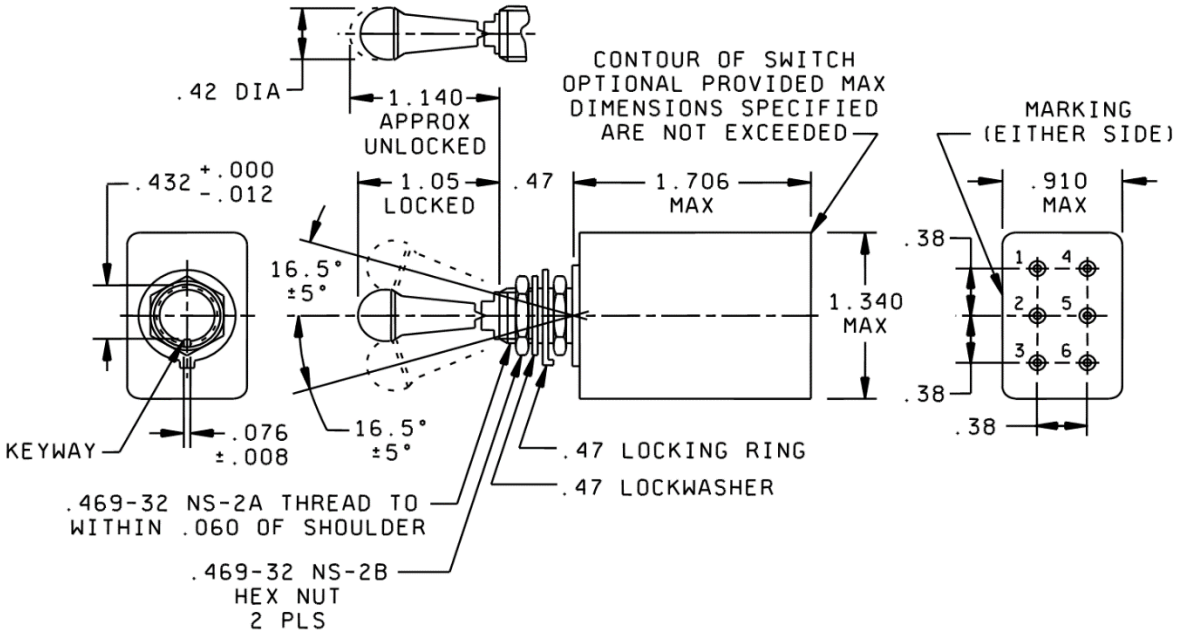
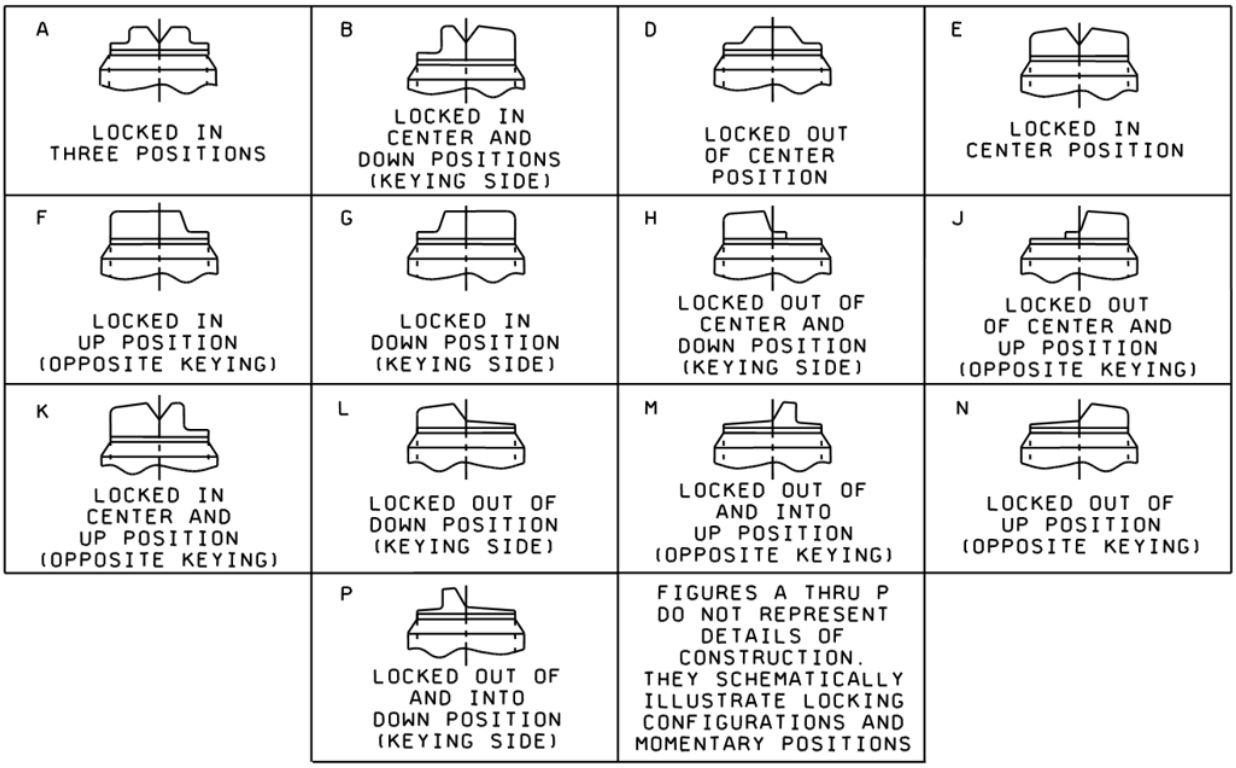


FIGURE1. Dimensions and configuration.

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LOCKING COMBINATIONS



NOTES:

1. Dimensions are inches.
2. Unless otherwise specified, tolerance is ± .020 for two place decimals and ± .005 for three place decimals.
3. For hardware detail specifications see appendix of MIL-DTL-3950.
4. Contact installing/ removal tool in accordance with M81969/14-03 per SAE-AS81969/14.
5. Grommet sealing plug, in accordance with MS27488-16-2 per MS27488.
6. Terminal shall adequately accept a wire contact within dimensional limits of M39029/1-102. per SAE-AS39029/1.
7. Part number example: MS27781-21A (locking combination ‘A’).

FIGURE1. Dimensions and configuration Continued

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REQUIREMENTS

Sealing plugs may be used in nonfunctional grommet holes.

Sealing grommet shall seal on a smooth wore insulation of .068 to 0.103 diameters

The terminal end of switch shall be color coded blue to indicate contact size

Direction of internal mechanism movement is opposite to the direction of the toggle movement

Maximum weight is .137 pound.

The locking means at the top of the toggle bushing shall be capable of withstanding a torque of 20 inch-pounds applied in both directions immediately following the humidity test.

Locking arrangement Positive locking shall be accomplished and shall prevent motion of the toggle lever until the locking mechanism shall be 3 to 5 pounds

Direction of internal mechanism movement is opposite to the direction of the toggle movement.

Electrical rating: See table I

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TABLE I Detail Requirements

MS Part Number	Available Locking combinations	Circuit with toggle lever in			Current capacity (amperes) 28 volts dc			Current capacity (amperes) 115 volts, 60 and 400 hertz ac		
		Keying side	Center	Opposite Keying side	Lamp Load circuit	Resistive circuit	Inductive circuit	Lamp Load circuit	Resistive circuit	Inductive circuit
MS27782-1	ALL	1-2 ON 4-5	1-2 ON 5-6	2-3 ON 5-6	7	20	15	4	15	15
MS27782-2	E,F,K,L,M,N	MOM 1-2 ON 4-5	1-2 ON 5-6	2-3 ON 5-6	5	18	10	2	11	8
MS27782-3	E,L,N	MOM 1-2 ON 4-5	1-2 ON 5-6	MOM 2-3 ON 5-6						
MS27782-21	ALL	1-2 ON 4-5	OFF	2-3 ON 5-6	7	20	15	4	15	15
MS27782-22	D,F,G	OFF	NONE	2-3 ON 5-6						
MS27782-23	D,F,G	1-2 ON 4-5	NONE	2-3 NO 5-6						
MS27782-24	E,F,K,M	NONE	OFF	2-3 ON 5-6	5	18	10	2	11	8
MS27785-25	F	NONE	MOM. OFF	2-3 ON 5-6						
MS27782-26	F	MOM 1-2 ON 4-5	NONE	MOM 2-3 ON 5-6						
MS27782-27	E,L,N	MOM 1-2 ON 4-5	OFF							
MS27782-28	E	MOM OFFf	OFF	NONE						
MS27782-29	F	MOM 1-2 ON 4-5	NONE	2-3 ON 5-6						
MS27782-30	F	MOM 1-2 ON 4-5	NONE	OFF						
MS27782-31	E,F,K,L,M,N	MOM 1-2 ON 4-5	OFF	2-3 ON 5-6						
MS27782-32	E	NONE	1-2 ON 4-5	MOM 2-3 ON 5-6						
MS27782-33	E,F,K,M	NONE	1-2 ON 4-5	2-3 ON 5-6	7	20	15	4	15	15

Test Requirements

Test shall be performed in accordance with MIL-DTL-3950 except

a) During all tests. Switches shall be fully wired with appropriate wire and terminal contacts.
b) Contact voltage drop- the contact drop with two terminals and the switch contact in series shall not exceed 8 millivolts measured from one wire contact through the contacts to the other wire contacts.

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Referenced documents

MIL-DTL-3950
MS27488
SAE-AS39029/1
SAE-AS81969/14

Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army – CR
Navy – AS
Air Force – 85
DLA – CC

Preparing activity
DLA – CC

(Project 5930-2024-020)

Review activities
Army – AR, AV, MI
Navy – EC, MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/ .

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